ISUZU EUROPE · AFTERSALES ANALYTICS · INTERNAL WORKING PAPER

How we sort 165,000 parts — identity on the truck, competition beside it

The parts classification behind our pricing and category reporting assigns every part three labels from its price-list description. This paper explains how that model works, how it checks against customs codes and Isuzu's own figure numbers, why one fused label cannot carry both what a part is and how it competes, and how the locked two-axis design compares with Volvo, Isuzu's catalogue, and the standards the aftermarket actually runs on.

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17 September 2026
Prepared for internal review

€50.71M
Billed parts examined (18 months, Jan 2025 – Jun 2026)

164,667
Classified part numbers

3 independent checks
Customs codes · Isuzu figures · part text

62.7%
agreement with customs commodity codes, on part count

71.4%
agreement with Isuzu's own catalogue figure groups

€1.84M
billed parts that were in a physically unrelated family

Two axes
identity live on the file; commercial score still to build

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01

What the model actually is

Every part we sell carries three labels: a commercial family, a system group, and a part type. All three were written by a keyword rule reading the part's description text in the price list. Nothing else feeds them.

The source workbook says so itself, on its summary sheet: *"3-level classification summary. Generated from Description field using rule-based mapping. Rows without a clear pattern are assigned to Other / review."* That is a legitimate way to start a classification, and it was executed cleanly — across 16,210 distinct descriptions there is not one case where the same text was given two different answers.

The consequence is equally simple: **the model can only know what the words say.** `FILTER; ACL` does not contain the word "air", so no rule reading that text can know it is an engine air cleaner. That single blind spot is the largest finding in this paper.

Layer 1

The source data

IE40 PR00 price list (SAP), 13 March 2026 — 164,380 rows. Each row carries the part number, a short description, the list price, a customs commodity code, and Isuzu's own parts-catalogue figure number.

Only one of those five fields is used to classify: the description.

WHAT CAN GO WRONG HERE

Description text is abbreviated trade shorthand, and a handful of strings in the live master are corrupted in place (`k0109AMP UNIT` was `HEADLAMP UNIT`). Around ten rows, small money, but this file is the source of truth for part text.

Layer 2

The method

Rule-based keyword mapping over the description, applied once in a workbook, then frozen. Rows with no clear pattern are sent to `Other / review`.

Fully deterministic: identical text always produces an identical answer. That was tested and it passes on all 16,210 distinct descriptions.

WHAT CAN GO WRONG HERE

One word can name two unrelated things. A door window *regulator* and an alternator voltage *regulator* are the same token; 107 part numbers were filed by the wrong meaning. And an abbreviation the rule has never been taught (`ACL` for air cleaner) is invisible.

Layer 3

The three levels

Level 1 commercial family (7 values) → **Level 2** system group (25) → **Level 3** part type (171 leaves). Example: `Competitive parts > Filters > Air filter`.

Levels 2 and 3 describe what the part is. Level 1 describes how we think it competes.

WHAT CAN GO WRONG HERE

A commercial judgement is fused into the part's identity. Competitive exposure depends on the market, not on the part's name — the same brake disc is contested in Germany and captive in Armenia. One label per part number cannot carry that.

Layer 4

The derived files

The workbook is turned into `pr00_classification.csv`, then `aim_pn_classification.csv`, then a second version with a safety gate that checks keys, vocabulary, hierarchy and drift.

This is what every dashboard and pricing study actually reads.

WHAT CAN GO WRONG HERE

Two build defects: part numbers are stored zero-stripped in one file and zero-padded in the next, so 1,653 rows (€870,378) were never covered by the gate's headline check; and 19.5% of keys take their labels from a superseding part number rather than their own row.

Layer 5

What reads it

The price revision analysis, the category dashboards, the distributor category cut, the opportunity ranking — and indirectly the service-capture view.

Service capture is the one consumer that does *not* depend on this tree; it reads Isuzu's figure numbers instead. That is why a figure-anchored identity view stays usable even when a commercial parent would hide a part.

WHAT CAN GO WRONG HERE

A classification reading becomes a business conclusion. If a wear or filter leaf looks empty, check whether the parts sit under a different commercial parent before calling a market a zero.

Chain verified by replaying the build scripts; row counts from the files themselves, 17 September 2026.

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02 The three levels, and the one that is not like the others

Levels 2 and 3 answer "what is this part". Level 1 answers "how does it compete". Mixing those two questions into one hierarchy is the design decision that everything else in this paper pushes against.

LEVEL 1

7
Commercial family

- Competitive parts
- Repair parts
- Accident parts
- High value repair
- Comfort parts
- Hardware & consumables
- Other / review

> **Tested and weak.** Against our own independent-brand price scrapes, parts labelled “Competitive” were 1.43× more likely to have an aftermarket rival than parts labelled anything else — but of the parts where a rival was actually confirmed, only 48.2% carry the Competitive label.

LEVEL 2

25
System group

- Brake parts, Filters, Exhaust
- Steering, Suspension, Cooling
- Clutch parts, Fuel system
- Body parts, Lighting, Mirrors
- … plus Hardware and two residual bins

> **Mostly real, three exceptions.** Belts, Mirrors and Brake parts are tight, coherent families. `Hardware`, `Hydraulic / mechanical` and `Electrical sensors` spread across 83, 65 and 103 customs subheadings respectively — 44,463 part numbers, 27% of the catalogue, in bins with a family name on them.

LEVEL 3

171

Part type (the leaf)

- Air filter, Oil filter, Fuel filter
- Brake disc, Brake drum, Brake pad
- Bumper, Door, Sensor…
- and one “Other …” leaf per family

> **The count is fine; the grain is not.** 171 leaves is the same order as the industry’s own commercial rollups (see section 6). The problem is that a leaf carries no attributes, so `Brake disc` holds 475 parts and one of them is 48.5% of the money.

THE CORE DESIGN QUESTION

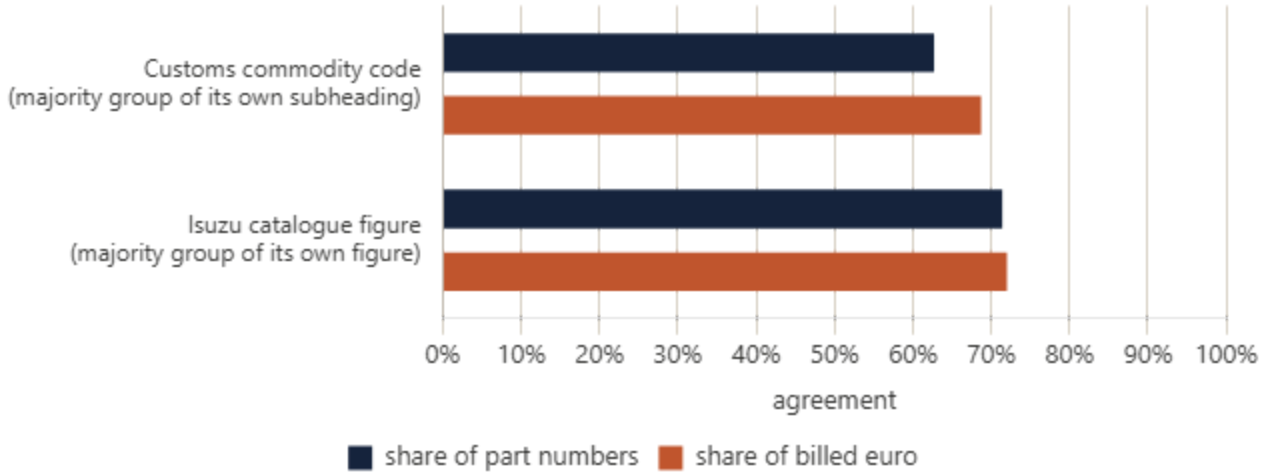
Competitive exposure is a property of a *market*: whether an independent brand offers that part, for that vehicle, in that country, at what price. It is not a property of a part’s noun. Every external classification we compared against — Isuzu’s own figures, Volvo, TecDoc, the US standard — keeps identity and commercial rollup as separate layers. We fused them, which is why a king pin kit (a steering wear item with real money on it) is labelled “Hardware & consumables” and disappears from every wear-item view.

03 How we tested it against sources it never saw

A keyword classifier cannot be validated by re-reading the keywords. So the tree was checked three times against data the rule had no access to: the customs code on each part, Isuzu’s own catalogue figure, and the part text read the way a parts person reads it.

Does the classification agree with sources it never read?

Share of parts — and of billed euro — sitting in the same group as their own customs peers, and in the majority group of their own Isuzu catalogue figure. 18-month window.



Neither source is a gold standard. Customs code `870899` ("other parts of motor vehicles") alone holds 15,780 of our parts and carries no functional meaning, and an Isuzu figure legitimately contains the fasteners of the assembly it illustrates. Read these as coherence, not as an error rate. What matters is that no check of this kind runs in the build today.

The third check: reading the part text like a parts person

This is the test that found the money. Twenty-two high-precision rules, each a phrase that names a part's system beyond argument, each with explicit exceptions so a brake *lamp switch* is not counted as a brake and an air-cleaner *bracket* is not counted as a filter. A violation is something a parts person would recognise on sight.

Result: 3,546 part numbers where the text rule and the catalogue place disagree — €2,091,184 billed, 4.12% of the 18-month book. That splits into two very different things, and they must never be added into one headline.

(a) Wrong system — the part is in a family it has no physical relation to

Service class	Part numbers	Billed € (18m)	Where a noun-only rule can put it
Engine air filtration	128	1,160,612	`FILTER; ACL` → Air conditioning / Other air system
King pin kits	536	219,775	`KING PIN KIT` → Hardware > Pin
Steering drag links	235	171,854	`LINK; DRAG` → Hydraulic / mechanical > Linkage
Bumper bars	201	148,994	`BAR; FRT BUMPER` → Suspension > Leaf spring
Control-arm ball joints	67	126,243	`JOINT ASM; BALL` → Hydraulic / mechanical > Linkage
Door window regulators	107	16,556	`REGULATOR ASM; DOOR` → Starter / alternator > Voltage regulator
Total wrong system	**1,274**	**1,844,034**	**live file places these under their truck identity**

Every one of these six is independently confirmed against Isuzu's own electronic parts catalogue: the catalogue's figure name for `FILTER; ACL` reads **1-30 Air Cleaner**, for the door regulator **6-31 Front Door Glass and Regulator**, for the king pin **4-10 Front Axle**. The parts catalogue and the part text agree on identity; a commercial or hardware parent is the wrong hang-from for a wear or system view.

(b) Object type versus system of use — defensible either way, applied inconsistently

2,272 parts and €247,150 sit on a decision the tree never made. 9,637 gasket-type parts are filed as `Gaskets / seals`; another 1,423 are filed under the system they serve (`GASKET; DPD` → Exhaust). 6,607 harnesses go to `Electrical sensors`; 673 go to their system. Neither choice is wrong. Having both in one tree is. This one is a decision to take, not a bug to fix — and section 6 shows a standards body has already written down the rule.

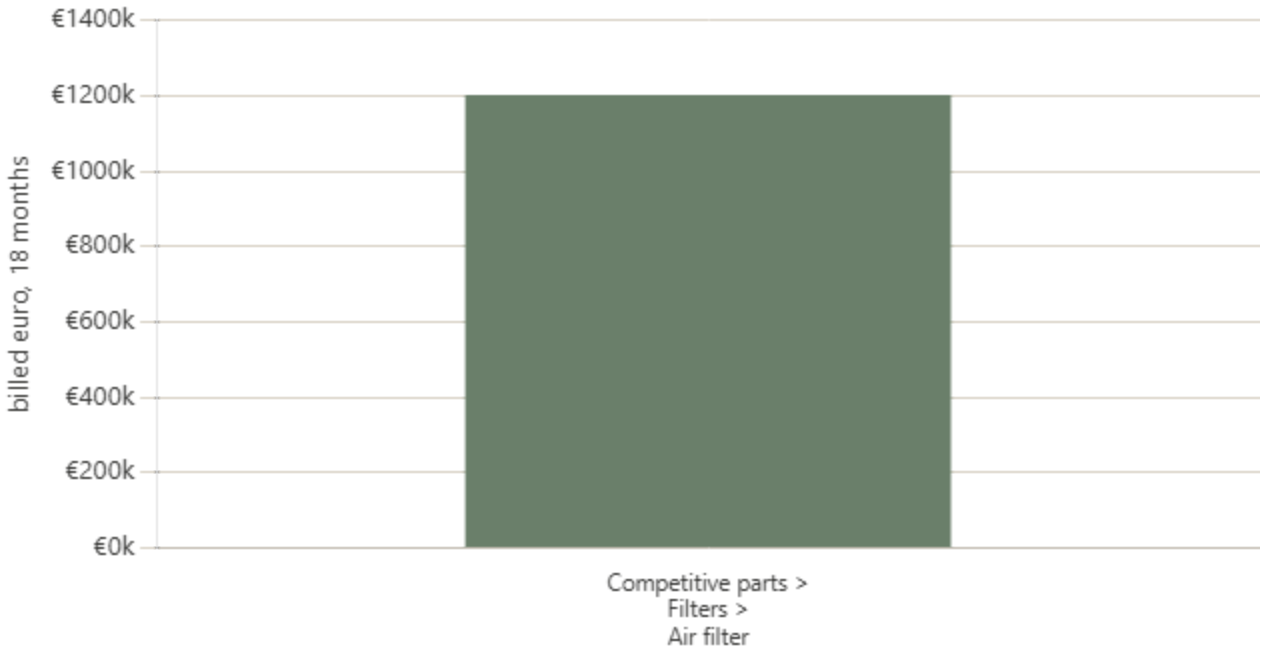
04 Engine air filtration as an identity example

Engine air filtration is €1.20M of billed parts over 18 months across 256 part numbers. On the live file they sit under Filters as air-cleaner identity — not under a comfort or hardware parent. The abbreviation on the price list is `FILTER; ACL`; Isuzu's figure for those rows is **1-30 Air Cleaner**.

That is the identity lesson in one leaf: when Axis A hangs from the figure or from Level 2 / 3, a wear or service view can still find the part. If the same row only hangs from a commercial parent that was assigned by reading the noun, a filter leaf can look empty while the euro is still being billed elsewhere in the tree.

Where engine air-filtration euro sits on the live identity cut

All 256 part numbers whose text reads as an engine air-cleaner element. Billed euro, 18 months to June 2026.



Cabin and blower filtration is a separate €1,622,097 across 147 parts and genuinely does belong under air conditioning. It is not mixed into this chart.

Euro, family and fleet together, so the number can be judged

A part count on its own says nothing — 61,243 units could be engines or bolts. Read against the fleet it serves, engine air filtration runs at 0.37 filters per truck per year across the

network, against a service interval that should put it far higher. That gap is the business finding.

Market	Trucks in service 2025	Billed €	Units	€ / truck	Units / truck
UTI (Israel, CV)	10,336	228,545	12,257	22.1	1.186
ITUK	22,389	217,383	7,734	9.7	0.345
IUK	47,783	177,863	10,187	3.7	0.213
Belgium	—	71,028	3,799	—	—
Norway	—	59,170	3,102	—	—
Ireland	—	52,605	2,267	—	—
Sweden	—	45,807	2,447	—	—
UMI (Israel, LCV)	18,542	32,545	2,838	1.8	0.153
Network	**163,459**	**1,203,146**	**61,243**	**7.3**	**0.370**

UTI and UMI are two separate Israel distributors with different fleets and are never blended. Markets without a trucks-in-service count carry euro only — no rate is struck for them.

> **WHY FIGURE IDENTITY MATTERS**
>
> **The service-capture view reads Isuzu figure numbers, not the commercial parent.** PMS (the scheduled preventive-maintenance service programme) derives its service category from the figure key. On this leaf, 207 of the 256 parts are tagged as air filters on the figure layer, covering €1,202,903 of the €1,203,146 — 99.98% of the euro. That is Axis A in practice: hang a wear or service view from the figure or from Level 2 / 3, not from a commercial box.

> ### Same mechanism, different part: king pin
>
> A `king_pin` service rule that only fires on parts under Suspension will see an empty set if every king-pin kit hangs under Hardware because the noun read “pin”. Under Axis A, those 536 part

numbers hang from front-axle / steering identity (figure **4-10 Front Axle**), so a wear view still finds the €219,775 billed in 18 months. Section 8 works this example next to the commercial-score axis.

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05 Four classifications, one part number, four answers

This is the finding that reframes the rest. The tree in this paper is not the only parts classification in the workspace — it is one of four, all live, all built differently, and no document says which one wins.

Worked example

part number `8980913940`, description `FILTER; ACL` — an engine air-cleaner element — on the live files:

1

The price-list tree

7 / 25 / 171 · 164,667 parts

Built by keyword rules on the description. This is the tree every pricing and category view reads.

> Live identity: **Filters > Air filter**

2

The category workbook

21 system names · 16,499 parts, ~79% of euro

A second vocabulary living in the same file, joined by description. Uses system names at the top level, not commercial families.

> System name follows the same identity cut when the description is clear

3

Isuzu's figure numbers

297 figures · feeds service capture

Derived from the parts-catalogue illustration number — the manufacturer's own assembly structure, not our words.

> Figure **1-30 Air Cleaner** — identity without reading the noun

4

The March 2026 functional workbook

15 numeric families · 15,353 billed parts · not wired in

An independent build with a functional top level and a full commercial segmentation (revenue contribution × sales velocity × margin tier).

> Node **05.02 Air Filter / Cleaner**

Four trees can still disagree on other leaves. The functional workbook should not simply be adopted wholesale: its own second level appears to have been sorted by description and filled downwards, so `01.01 Cylinder Block & Crankshaft` holds 312 parts of which 145 are simply called "BEARING", and at least 59 of those are provably gearbox or axle bearings.

Each tree is right where the others are thin. That is the argument for written precedence (which classification wins, for what purpose) rather than assuming one file answers every question. Axis A on the live classification is the identity contract for pricing and category views; figure numbers remain the hang-from for service capture.

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06 How the industry does it

Four public reference points: a commercial-vehicle OEM's own catalogue (Volvo), Isuzu's own published scheme, the European aftermarket standard (TecDoc), and the US standard (ACES / PIES). They agree with each other on two things, and both of them are things we do differently.

Volvo Trucks — function groups 0–9

Volvo Impact: ten top groups, then a two-digit level (`51`), then a three-digit level (`511`) carrying the chassis-specific detail.

0	General
1	**Service and maintenance**
2	Engine, mounting and equipment
3	Electrical system and instruments
4	Power transmission
5	Brakes
6	Wheel suspension and steering
7	Frame, springs, damping and wheels
8	Body, cab and interior
9	Miscellaneous

Two things stand out against ours. Volvo's top level is purely functional — it says where the part is on the truck, never how it competes. And **"service and maintenance" is its own group**, so scheduled-service parts are addressable as one set. Ours are scattered across Filters, Air conditioning and Air system, which is exactly why service parts must be addressable as one set. Scania uses the same idea with two-digit function groups; Mack runs a function-group system too, but its group list is not public.

Isuzu's own scheme — figure numbers `X-XX`

Published in every parts catalogue: the first digit is the section, the next two the subsection. `0-15` = Engine (0), Crankshaft / Piston / Flywheel (15).

0	Engine
1	Intake, cooling, fuel, exhaust — where `1-30 Air Cleaner` lives
2	Clutch and transmission
3	Brakes
4	Propeller shaft, axles, steering, suspension, wheels
5	Frame and cab mounting
6	Cab and body panels
7	Trim, mirrors, instrument panel, seats
8	Electrical, heater and air conditioning
9	Tools

The live price list carries **297 distinct figure numbers**, and every part already has one. This is the cheapest independent check available to us — it is on the same row as the description, it was never read by the classifier, and it is the layer that kept the service-capture numbers aligned with figure identity rather than a commercial parent.

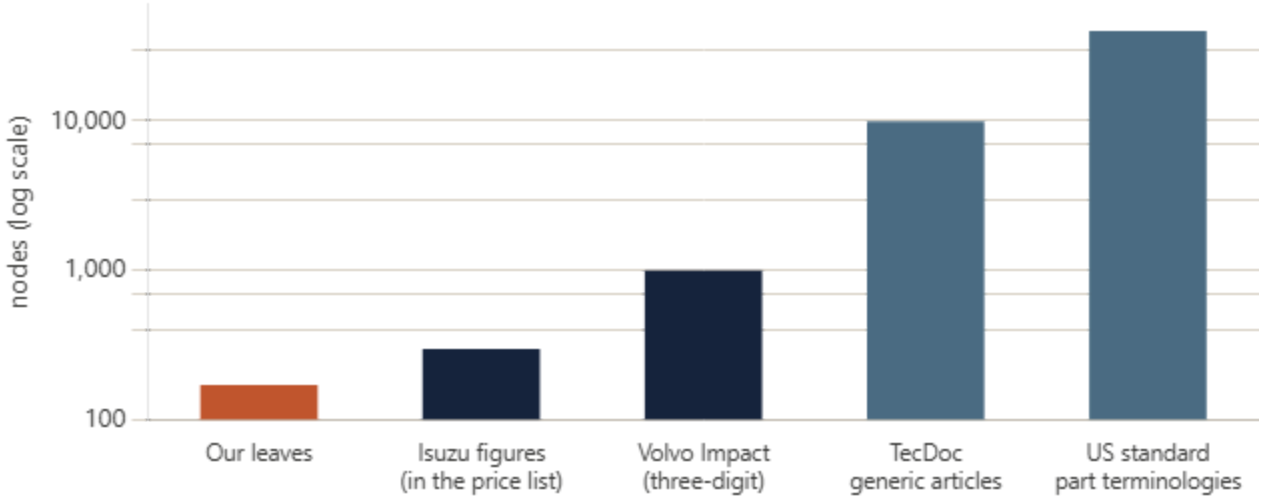
The two standards the aftermarket runs on

TecDoc (TecAlliance) is the European de facto standard. Parts are classified to *generic articles* — a five-digit number with a standardised designation, an assembly group, a purpose of use, and typed criteria marked mandatory, permitted or forbidden. The number that matters for us: TecAlliance rolls its **~9,900 generic articles up into 32 main product groups and 160 sub-categories** for commercial sales reporting to the international trading groups. A functional standard at the bottom, a commercial rollup on top, each free to change on its own schedule.

ACES / PIES (Auto Care Association) is the US equivalent. Its product classification database is a category → subcategory → part-terminology hierarchy with **40,000+ part terminologies**, and a separate attribute database adds **9,700+ attributes** assigned per terminology — up to 144 attributes on a single part type. Version 5.0 / 8.0 became official on 26 March 2026.

How fine is the identity grain, ours against theirs?

Number of distinct nodes at the identity level — our leaves against Isuzu figures, Volvo's three-digit groups, TecDoc generic articles and the US part terminologies. Log scale.



Note on an earlier reading of mine: I first compared our 171 leaves against TecDoc's 9,900 generic articles and concluded we were not in the same league. That was the wrong comparison — it puts our *commercial reporting* grain against their *fitment identification* grain. Against TecDoc's own commercial rollup of 32 groups and 160 sub-categories, our 25 groups and 171 leaves sit in a credible band.

The structures side by side

	Our tree	Isuzu figures	Volvo Impact	TecDoc	ACES / PIES
TOP LEVEL	7 commercial families	10 sections	10 function groups	32 product groups	categories
SECOND LEVEL	25 system groups	~30 subsections	~100 (two-digit)	160 sub-categories	subcategories
IDENTITY GRAIN	171 leaves	297 figures	~1,000 (three-digit)	~9,900 generic articles	40,000+ terminologies

ATTRIBUTES PER NODE				
Our tree none	Isuzu figures none	Volvo Impact —	TecDoc mandatory + permitted criteria	ACES / PIES 9,700+ attributes

TOP LEVEL IS				
Our tree commercial	Isuzu figures functional	Volvo Impact functional	TecDoc functional	ACES / PIES functional

SERVICE PARTS				
Our tree scattered	Isuzu figures by system	Volvo Impact own group (1)	TecDoc —	ACES / PIES —

What the comparison actually says

1. Nobody else puts a commercial judgement in the identity. Isuzu's figures, Volvo, TecDoc and the US standard all classify what the part *is*. TecDoc's commercial rollup is *derived from* the functional keys and allowed to move independently — TecAlliance's stated reason for building it that way is that suppliers then need no new implementation. We fused the two into one Level 1, which is why the "Competitive" label catches under half the competition we can already prove exists.

2. There is no attribute layer, and that is the real gap. TecDoc's criteria and the US attribute database are what make a category actionable: you can address every brake disc of a given diameter and thickness. Our leaves carry no attributes at all, which is why `Brake parts > Brake disc` is 475 parts with one of them holding 48.5% of the euro. That is a missing dimension, not a missing level — adding a fourth level would not fix it.

3. eCl@ss has already written down the rule we broke. In eCl@ss 16.0 (48,283 classes, published 28 November 2025) spare parts get a dedicated fourth-level "91" class which deliberately drops all product-describing properties, explicitly "to avoid that a certain product can be described as a distinct product class and as a parts-class for another parent product". That is a standards body stating the one-thing-one-home rule — the rule our gaskets, harnesses and pipes break in both directions today.

And one negative finding worth recording: there is no VDA or Odette standard for parts *grouping*. VDA 9004, 9007 and 9008 cover spare-part identification codes, logistics master

data and packaging exchange — identification and packaging, not taxonomy. In Europe, "compliant" in practice means mapped to TecDoc generic articles.

On the commercial axis specifically

There is no published named segment list for parts posture, so our Level-1 vocabulary is not deviating from a standard — there is no standard to deviate from. What does exist: **captive / semi-captive / commercial** is the working vocabulary in service-parts pricing practice, and it is *scored* from make-or-buy, competitor count, volume, supplier count, market share and lifecycle stage — much closer to the measured attribute we need than to a label read off a noun. McKinsey's current position argues directly against what we do: with catalogues above 100,000 parts, pricing at product-category level is the problem, and the recommendation is learned microsegments of parts with similar pricing power, clustered on customer, manufacturer and geographic parameters beyond the product itself, with 2–6% of sales as the reported uplift. The one legally fixed boundary in this space is EU Regulation 461/2010, which distinguishes "original spare parts" from "spare parts of matching quality" — and that is a property of the *competing* part, an input to exposure, not a label for our own part number.

Sources: Volvo Impact function-group documentation; Isuzu NPS81 / NQR81 and Giga CYZ parts catalogues ("How to refer to Parts Catalog"); TecAlliance myITG reporting structure, 9 May 2023, and the TecDoc Data Format specification (tables 320, 323, 324, 331–333); Auto Care Association product classification and attribute databases, ACES 5.0 / PIES 8.0 release 26 March 2026; eCl@ss 16.0 classification-class specification; Syncron product segmentation; McKinsey, *The future of aftermarket pricing*; EU Regulation 461/2010 as quoted in Case C-354/14. One trap for anyone re-checking these: TecAlliance's product pages advertise "360+ generic articles", which is the coverage of one data product, not the size of the standard.

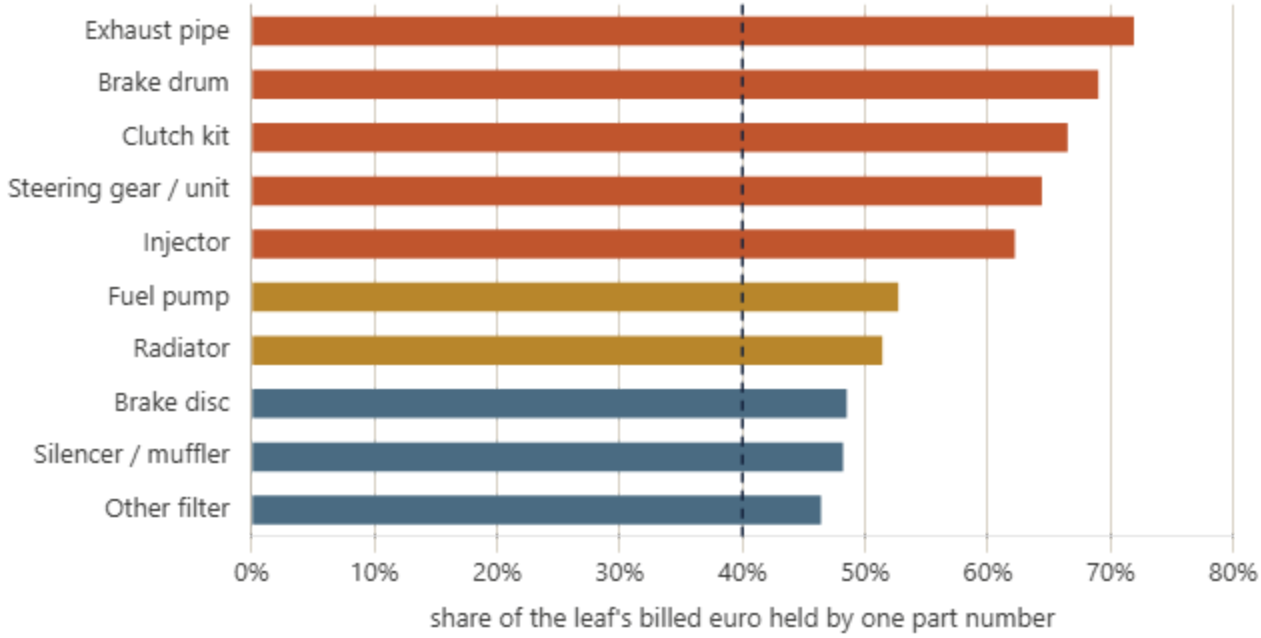
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07 Where the leaves are too coarse to price on

A leaf is usable when the parts inside it are commercially alike. Two symptoms say it is not: it mixes physically different goods, or one part number effectively *is* the leaf. **34 leaves carrying €30.9M — 65.1% of the 18-month book — fail on one or both.**

Which “families” are really one part number wearing a family name?

Largest part number as a share of its own leaf’s billed euro. Leaves above €100k. The dashed line marks 40% — above it, a leaf has stopped describing a family and is describing one product.



When someone says “brake discs rose 7.8%”, roughly half of that movement is one part number. Fifteen leaves are in this condition.

The other symptom is price spread inside a single leaf. `Body parts > Other body` runs from €0 to €16,615 with a median of €12 — a ratio of 1,435× between the top item and the middle of the leaf. `Exhaust > Aftertreatment` is 1,015×. A leaf holding both a €0 item and a €16,615 item is not a pricing unit.

How much of the book sits in a box that admits it is not a category?

Billed euro over 18 months, by how well the classification names it.



Nearly one euro in five is in “Other…”, “Review needed”, or no node at all. Every family has its own overflow bin and several of them are among the largest leaves in that family — `Air conditioning > Other A/C` at €1.11M was the biggest overflow bin in that family; engine air-cleaner identity now sits under Filters.

08 How the catalogue works now: two axes, then attributes

One label per part cannot carry both what the part is and how it competes. The locked design splits those jobs: Axis A is identity on the truck; Axis B is a commercial score beside it; attributes (size, position, duty) come later so a leaf can be priced as a comparable set.

Why one fused label is not enough

STRUCTURAL LIMIT

Level 1 as commercial family
Assigned by reading the part's noun. Same value in every market, for every customer, forever — so it cannot answer "contested here".

Levels 2 & 3 under that commercial parent
What the part is — but if a steering wear item hangs under Hardware, a wear-item view that starts from the commercial parent never sees it.

No attributes on the leaf
Nothing describes size, position, fitment or duty, so a leaf cannot be narrowed to a set you can actually price against.

The commercial "Competitive" box alone still under-catches rivals we can already prove from same-shop evidence; most of the book still sits in leaves too wide to treat as one price unit without further attributes.



Locked — two axes (17 Sep 2026)

JEROME DECISION — PROCEED

Axis A — identity only
What the part is and where it sits on the truck. Existing three levels (`catalog_l1` / `l2` / `l3`), figure-anchored. Wear views hang from Level 2 / 3 or the figure key — never from a commercial parent.

Axis B — commercial score, per part *per market*
Contested or not. A measured attribute beside Axis A — not the box identity hangs from. No fourth taxonomy layer.

Attribute layer — still open
Size, position, duty class. Explicitly out of this lock; needs its own call before any new levels.

Axis A is live on the classification files today. Axis B scoring is the next build on evidence we already hold (same-shop listings, Eucon, figure structure) — not a rename of Level 1.

Three short examples

1. King pin / steering wear

Filed under a hardware or commercial parent, a king-pin kit disappears from a wear view that starts at that parent. Under Axis A it hangs from Level 2 / 3 or figure **4-10 Front Axle**, so the wear view still sees the part and the billed euro.

2. Contested rivals

The commercial "Competitive" box alone catches only part of the rivals we can prove. Axis B scores contested / captive beside identity, per part per market, without moving the part off its truck system.

3. A leaf too wide to price

`Body parts > Other body` runs from a near-zero item to a five-figure panel with a median of about €12. Without size, position or duty on the row, that leaf is not a pricing unit — attributes are what make a comparable set, not another Level-1 rename.

What this lock settles, and what it leaves open:

#	Item	Type	Status
1	Axis A identity live (`catalog_l1` / `l2` / `l3`, figure-anchored)	Locked	**Live on file**
2	Wear / service views hang from Level 2 / 3 or figure — never from commercial Level 1	Locked	**Contract**
3	Written precedence: which classification wins, for what purpose	Locked	**Recorded**
4	Axis B commercial score per part per market	Build	Next build
5	Attribute layer (size, position, duty) before any new taxonomy levels	Decision	Needs your call

#	Item	Type	Status
6	Object-vs-system home for gaskets, harnesses and pipes	Decision	Needs your call

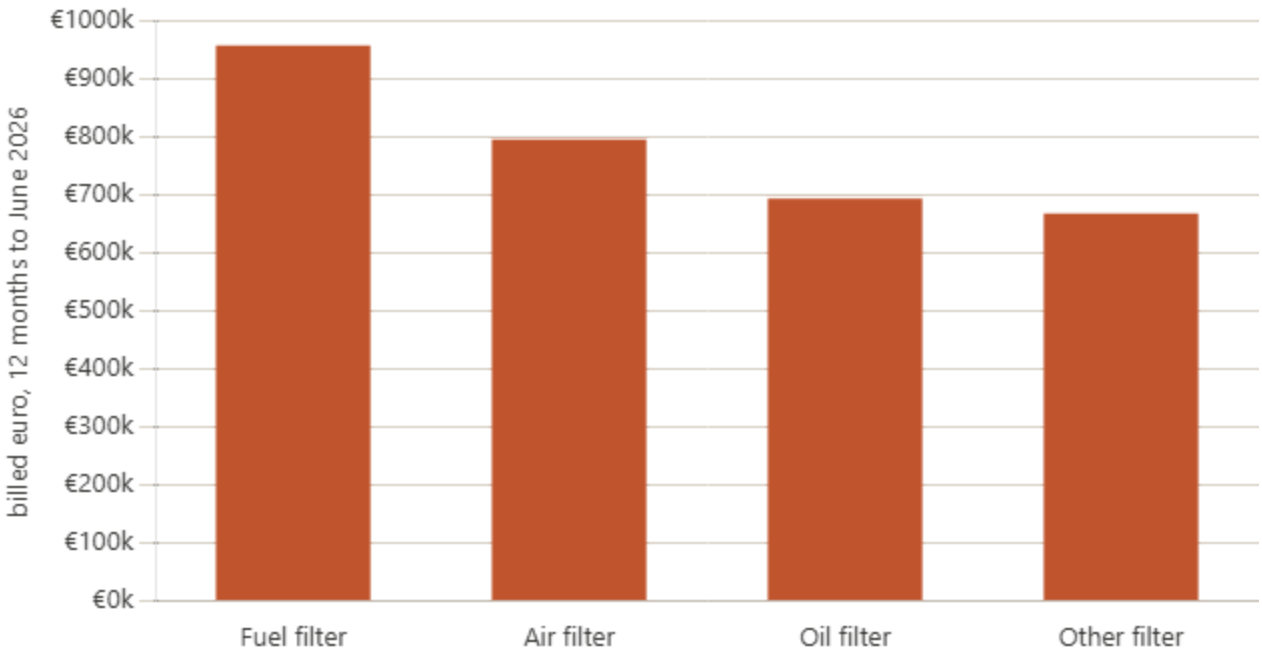
09

What is still open

Axis A identity is live. The items below are the remaining product decisions and builds — not a backlog of finished safety work.

How the Filters family reads on the live file

Billed euro in the 12-month in-scope window used for the price revision (July 2025 – June 2026), by filter type. Air filter is a real category on this cut: **€796,783 across 40 part numbers** — 25.5% of the Filters group, second only to fuel filters. Largest markets: UTI €159,028, ITUK €133,147, IUK €114,287.



The Filters group total is the same billed euro partitioned by identity. Nothing here changes a list price; it changes which family the money is counted in.

Open items, in order of cost to leave open

Open item	Exposure	Why it matters
"Other / review" residual after the high-precision pass	~9.5k parts · ~€1.75M billed in the residual bin	Further clear-out needs figure-key majority or a new leaf decision — not more text rules.
Axis B commercial score not yet scored on the file	network	Two axes are locked; the contested / captive measurement per part per market still has to be built beside Axis A.
EQ-Hit column still empty on the classification file	rule classes confirmed; column not filled	Systematic fill is what turns identity checks into a measured build column.
No attribute layer	34 leaves · €30.9M share of book	Leaves that carry most of the book are too dispersed to be pricing units without size / position / duty. Explicitly out of the two-axes lock.
Object-vs-system for gaskets, harnesses and pipes	2,272 parts · €247k	Needs one explicit call; both homes exist today.

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If you are working on the price revision

What to use, and what not to trust

This classification is safe for mix and category reporting when you respect the two-axes contract below. The safety gate must PASS before quoting a family figure.

USE

- `aim_pn_classification_v2.csv` — Axis A identity columns, two-axes contract in the meta file.
- `aim_catalog_focus_filters.csv` for the filter category cut. Take the **catalog_full** basis, not the restricted one — the restriction drops nearly nine tenths of the euro.

- Isuzu figure numbers when you need a part's position on the truck. They are on every row and they are the hang-from for service capture.
- The service-capture files as they stand — they read figure numbers, not the commercial Level-1 parent.

DO NOT TRUST

- The Level-1 commercial family as a measure of competitive exposure. It is a noun-derived label, not a market fact. Contested / captive is Axis B, beside identity.
- A leaf as a pricing unit when one part number is most of it — check concentration first (section 7).
- Any category euro that ignores Axis A identity (for example reading a wear view only from a commercial parent).

The live classification is a partition of the same billed euro: the network weighted base-price movement ties to the shipped comparison exactly (5.9616% against 5.9616%, step euro 1,652,140.71 against 1,652,140.71), and the per-distributor tie-out passes. Identity placement does not change a list price; it changes which family the money is counted in.

Footnote: category cuts printed before 17 September 2026 may still show engine air filtration under an air-conditioning overflow bin. Use the live filter cut above.

Israel: UTI and UMI are two distributors and are reported as two rows throughout this paper — never blended. Figures on the live classification as of 17 September 2026.

Method

Every figure in this paper was computed from the workspace data, not copied from an earlier document. Sales window for euro figures: 1 January 2025 to 30 June 2026 (18 months, gross billed, €50.71M on classified part numbers) — except section 9's price-revision cut, which uses the 12-month in-scope window July 2025 to June 2026. The two bases are not mixed.

Three independent checks: customs commodity code from the price master; Isuzu parts-catalogue figure number; and 22 high-precision description rules with explicit exceptions. Identity placements for the high-precision classes were confirmed against Isuzu's electronic parts catalogue.

Underlying files

- `00_docs/CLASSIFICATION_FULL_AUDIT_20260917.md` — the full audit this paper summarises
- `01_data/00_derived/aim_pn_classification_v2.csv` — the classification
- `01_data/00_derived/aim_classification_gate.json` — safety gate result
- `01_data/00_derived/_audit_cls_description_violations.csv` — text-rule vs catalogue disagreements, line by line
- `01_data/00_derived/aim_catalog_focus_filters.csv` — the live filter cut
- `01_data/90_INBOX/_patch_cls_eq_confirmed_20260917_report.json` — identity placement report

Terms used once and then abbreviated

- **PMS** — the scheduled preventive-maintenance service programme
- **Figure number** — Isuzu's own parts-catalogue illustration group, in the form `X-XX`
- **HS code** — the customs tariff commodity code carried on the price list
- **Generic article** — TecDoc's unit of part identity
- **Trucks in service** — the vehicles-in-operation count used as the fleet denominator